

Beyond the Shadow of a Doubt: Benchmarking MCMC Sampling Algorithms for Multimodal Astrophysical Inference

Progress Report

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April 13, 2026

Abstract

As astrophysics enters the era of petabyte-scale datasets, the choice of inference algorithm and implementation for a research problem could make the difference between an intractable or incorrect analysis and a major scientific discovery. The high dimensionality and multi-modality of posterior distributions can pose problems for popular inference tools. To ensure that the astrophysics community is fully leveraging available inference algorithms, we benchmark a set of widespread tools from across the astrophysical literature. We benchmark seven inference algorithms—random walk Metropolis-Hastings (RWMH), affine-invariant ensemble sampling, parallel tempering, NUTS, SMC with tempering, non-reversible parallel tempering, and nested sampling (as a non-MCMC comparator)—on two test problems: a synthetic 8-component 2D Gaussian mixture with known ground truth, and a simulated exoplanet transit with stellar spots. All experiments run on AMD EPYC 9474F 48-core CPUs and Nvidia L40S GPUs using JAX to ensure implementation-agnostic comparison. We evaluate each sampler on mode recovery, posterior accuracy, and computational efficiency (ESS per log-likelihood evaluation and ESS per wall-clock second), and will produce a concrete set of recommendations for practitioners.

1 Project Overview

The central question of this project is: *Which parallelizable MCMC-family samplers most reliably recover multi-modal posteriors in astrophysical settings, and at what computational cost?* We aim to benchmark seven algorithms: Random-Walk Metropolis Hastings (RWMH), affine-invariant ensemble sampling [1, 2], reversible parallel tempering (Stochastic Even Odd, SEO) [3], non-reversible parallel tempering (Deterministic Even Odd, DEO) [4], NUTS, SMC with tempering [5], and nested sampling [6] as a non-MCMC reference.¹

We evaluate each sampler on two test problems. The first is an 8-component 2D Gaussian mixture with fully known ground-truth modes, weights, and covariances, enabling precise measurement of mode recovery and posterior accuracy. This problem is advantageous for two reasons: its simple analytical form and low-dimension provides an intuitive visualization of each sampler’s results, yet, despite this simplicity, accurately recovering all eight components remains a challenging task. The second is a simulated exoplanet transit with active regions on the host star’s surface, a problem known to produce multi-modal and highly correlated posteriors [?, ?, ?]. Active regions have been a growing source of bias in exoplanet science [?], yet no systematic benchmarking has identified the most reliable sampler for jointly constraining active region and exoplanet properties. The evaluation criteria are discussed in Section 6.

The full implementation stack is JAX² [8] / BlackJAX³ [9] / NumPyro⁴ [10], ensuring hardware-accelerated, implementation-agnostic comparison across all algorithms.

¹Ensemble Kalman sampling [7] was originally planned as an eighth algorithm but is deferred pending completion of all other sampler runs; it will be included if time permits.

²<https://github.com/jax-ml/jax>

³<https://github.com/blackjax-devs/blackjax>

⁴<https://github.com/pyro-ppl/numpyro>

2 Division of Labor Update

The division of labor has been reshuffled relative to the original proposal. Lucas Arthur has taken ownership of both parallel tempering variants—reversible SEO and non-reversible DEO—in addition to his originally assigned algorithms (NUTS and SMC with tempering).

In addition to his originally assigned algorithms (RWMH, affine-invariant MCMC, and nested sampling), Samson Mercier has taken on minor development of the forward transit model.

The ensemble Kalman sampler (EKS) is de-prioritized. Implementation is on hold until all other samplers have been run and results collated across both test problems. Samson Mercier will implement it if time permits; it will be omitted from the final comparison if it is not ready in time.

3 Progress to Date

3.1 Gaussian Mixture Model — Substantially Complete

Seven of the eight samplers have been fully implemented and run on the Gaussian mixture problem: RWMH, affine-invariant MCMC, reversible PT (SEO), non-reversible PT (DEO), NUTS, SMC with tempering, and nested sampling. EKS is the sole missing algorithm; it is deferred pending completion of transit model runs.

Each sampler produces the following outputs: posterior samples in a standardized format, corner plots, and a diagnostics record containing ESS, $\hat{R}$, acceptance rates, recovered mode weights, and inter-mode transition counts (for chain-based methods).

3.2 Stellar Spot Transit Model — Forward Model Complete, Sampling Not Yet Started

The transit forward model has been implemented using the **SAJAX** library⁵ with a NumPyro likelihood and BlackJAX-compatible log-density interface.

Each active region is described by four parameters (longitude, latitude, angular size, and contrast); three stellar parameters (two limb-darkening coefficients and rotational period) and seven planetary orbital parameters complete the model, yielding 17 free parameters after fixing the mid-transit time (typically known to minute-level precision). Tight Gaussian priors are placed on the orbital and rotation periods, centered on truth.

Samson Mercier has validated the forward model against independent transit-fitting codes. Sampler scripts for the transit problem have not yet been written. Writing and running these scripts is the primary remaining task.

4 Preliminary Results: Gaussian Mixture

Notable observations include:

- **RWMH** shows poor mixing between modes. The chain rarely crosses the low-density region separating mixture components, resulting in substantially fewer inter-mode transitions and lower ESS per evaluation than all other methods. Performance is sensitive to initialization; chains started near a single mode frequently fail to discover remaining components.
- **Affine-invariant ensemble sampling** shows moderate inter-mode mixing when walkers are initialized broadly across the support, but degrades substantially when the ensemble is concentrated near a single mode.
- **Reversible PT (SEO)** achieves inter-mode mixing through temperature swaps, though at a lower rate than DEO at matched compute budgets.

⁵<https://github.com/SamMerc/sajax>

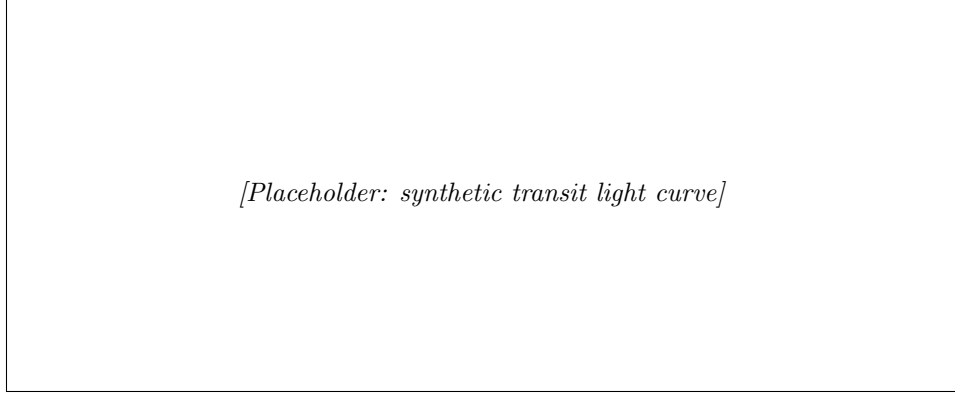


Figure 1: Synthetic exoplanet transit light curve generated by the **SAJAX** forward model. The x -axis shows time from mid-transit (hours) and the y -axis shows normalized stellar flux. The transit profile is modulated by two active regions—a dark spot and a bright facula—on the stellar surface.

- **Non-reversible PT (DEO)** achieves substantially more inter-mode transitions than reversible PT (SEO) at matched compute budgets, consistent with the theoretical prediction that non-reversible temperature swaps improve round-trip rates [4].
- **Nested sampling** reliably recovers all eight modes. Its ESS per log-density evaluation is lower than gradient-based methods, but it provides an accurate evidence estimate and serves as a robust reference.
- **NUTS** and **SMC with tempering** show strong ESS per evaluation on this low-dimensional problem; their relative performance on the higher-dimensional transit model remains to be seen.

Initialization sensitivity is a recurring theme across chain-based methods; we are actively exploring parameter tuning to establish fair comparison baselines ahead of the transit model runs.

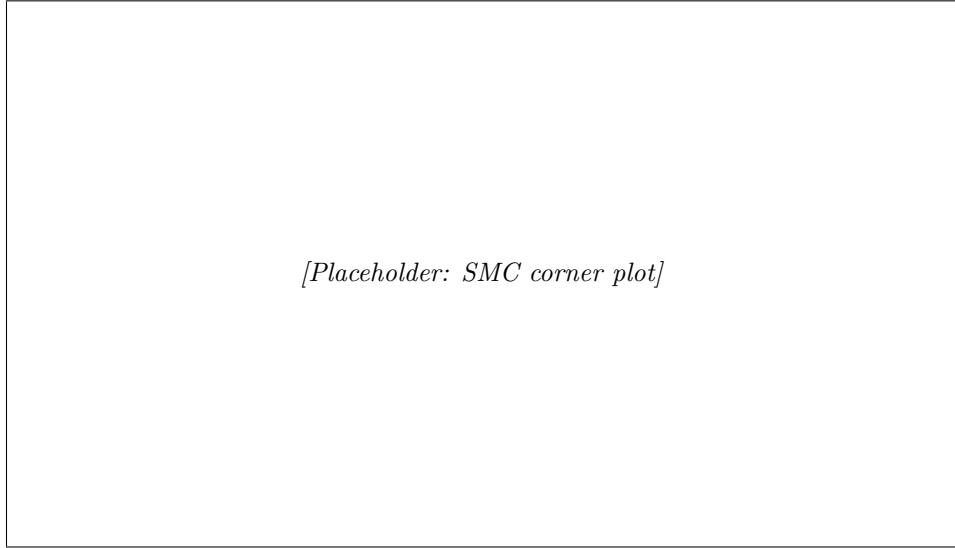


Figure 2: Corner plot of the SMC with tempering posterior on the 8-component 2D Gaussian mixture. Diagonal panels show 1D marginal posteriors with parameter value on the x -axis and normalized density on the y -axis; the off-diagonal panel shows the 2D joint marginal with x_1 on the x -axis and x_2 on the y -axis. Black crosses mark the true component means.

5 Revised Plan and Remaining Tasks

Table 1 lists the remaining tasks and internal deadlines.

Task	Owner	Target
Run all 7 implemented samplers on transit model	Both	Apr 25
Collate all results; build comparison tables	Both	Apr 25
Implement EKS (if time permits)	Samson	Apr 25
Produce final figures	Both	Apr 28
Write final report	Both	May 1

Table 1: Remaining tasks and internal deadlines.

6 Planned Figures and Evaluation Framework

Comparing the seven algorithms requires a unified evaluation framework because they produce fundamentally different outputs: correlated samples (RWMH, PT, NUTS), weighted particle sets (SMC, nested sampling), or continuously evolved ensembles (affine-invariant). We therefore report efficiency as ESS per log-likelihood evaluation and ESS per wall-clock second. The primary evaluation criteria are:

1. **Mode-recovery metrics:** the per-mode weight error $|\hat{w}_k - w_k|$, total variation distance from the true mixture, and (for chain methods) inter-mode transition count. For the Gaussian mixture, these are evaluated directly against ground truth; mode recovery is excluded from the transit model comparison since ground-truth modes are not readily available.
2. **Computational efficiency:** ESS per log-likelihood evaluation and ESS per wall-clock second. For chain-based methods, warm-up evaluations and warm-up time are included in the respective denominators.

Secondary diagnostics include the Gelman–Rubin potential scale reduction factor ($\hat{R}$), acceptance rates, and—for parallel tempering variants—swap acceptance rates and round-trip counts.

The main body will contain three figures:

1. **KDE posterior overlay (Gaussian mixture):** one panel per sampler, arranged in a grid. The x -axis is the first mixture dimension x_1 and the y -axis is the second dimension x_2 ; filled contours show the true mixture density and overlaid KDE contours show the sampler’s recovered posterior.
2. **Transit model comparison:** a two-panel corner plot contrasting the best- and worst-performing samplers on the transit problem. Along the diagonal, each panel shows a 1D marginal posterior with the parameter value on the x -axis and normalized density on the y -axis. Off-diagonal panels show 2D joint marginals with the two respective parameter values on the x - and y -axes, with contours at the 68th and 95th percentile levels.
3. **Summary table:** rows are samplers; columns are ESS per log-likelihood evaluation, ESS per wall-clock second (both reported for each test problem), fraction of modes recovered (Gaussian mixture only), $\hat{R}$, and acceptance rate.

Full per-sampler diagnostics are relegated to an appendix. For each sampler on both test problems, the appendix includes: a scatter plot of raw posterior samples (Gaussian mixture: x -axis x_1 , y -axis x_2 ; transit model: one 2D projection per parameter pair), trace plots for each parameter (x -axis: iteration number; y -axis: parameter value), and a full corner plot (diagonal panels: parameter value vs. normalized density; off-diagonal panels: pairwise parameter values with density contours).

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